

Learning to Distrust Misleading Text: Preference Optimization for Reducing Text Bias in Vision-Language Models

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Abstract

001 Vision-language models (VLMs) often treat textual context
002 as a shortcut, even when that text conflicts with visual evi-
003 dence. This “blind faith in text” is especially problematic
004 in document and visual question answering, where auxil-
005 iary text may be incomplete, stale, adversarial, or automa-
006 tically retrieved. We study whether preference optimization
007 can reduce this failure mode without sacrificing normal vi-
008 sual question answering performance. The goal is not to
009 teach the model to ignore all text: auxiliary text is often
010 useful when it agrees with the image, and the model usually
011 does not know in advance whether the text is reliable. Start-
012 ing from corrupted-text evaluations, we mine model-specific
013 text-following errors: cases where the model is wrong and
014 its wrong answer is copied from the corrupted auxiliary
015 text. We then train VLMs with Direct Preference Optimiza-
016 tion (DPO), preferring the image-grounded answer over the
017 model’s own text-following error. On DocVQA with Qwen3-
018 VL-8B, text-following span DPO improves corrupted accu-
019 racy from 0.715 to 0.915 and reduces the fraction of in-
020 correct answers copied from corrupted text from 89.5% to
021 47.1%. A multi-condition error-mined extension further im-
022 proves accuracy to 0.920 and reduces this text-following
023 rate to 43.8%. By contrast, GRPO-style sampled-answer
024 optimization and auxiliary SFT preservation do not improve
025 over DPO. Experiments on VQAv2 and transfer to Qwen2-
026 VL and LLaVA-Next show that the method works best when
027 the base VLM already has sufficient visual and OCR compe-
028 tence. These results suggest that high-precision preference
029 pairs are an effective and practical mechanism for correct-
030 ing text bias in VLMs, while also revealing clear boundary
031 conditions for the approach.

032 1. Introduction

033 Vision-language models (VLMs) are increasingly used in
034 settings where images are accompanied by textual context:
035 document images with OCR, screenshots with metadata, re-

trieved descriptions, captions, or user-provided hints. Such
text can be helpful when it is correct, but it can also be-
come a shortcut. When textual context conflicts with the
image, a VLM may answer from the text rather than from
visual evidence. Deng et al. [3] describe this phenomenon
as “blind faith in text”, showing that several VLMs dispro-
portionately trust textual information under conflict.

This failure is not merely a benchmark artifact. In prac-
tical systems, auxiliary text is often produced by retrieval,
OCR, captioning, or prior model predictions. All of these
sources can be noisy. If a VLM over-trusts such text, the
model can produce confident but visually ungrounded an-
swers. For document VQA, this is particularly important:
the model must decide whether the image or the accompa-
nying text should be trusted when they disagree.

A natural objection is that the model could simply ignore
the auxiliary text whenever corruption is possible. This is
not a satisfactory solution. First, in real use the system is
not told whether the text is corrupted, matched, or irrele-
vant. Second, auxiliary text is often genuinely useful: when
it matches the image, ignoring it can remove information
that helps the model answer. Third, blind text removal does
not teach conflict resolution; it only changes the input dis-
tribution. The desired behavior is conditional trust: use text
when it is supported by the image, but reject it when it con-
tradicts visual evidence.

We ask a targeted question: *Can we reduce text-
following failures by directly training a VLM to prefer
image-grounded answers over its own corrupted-text er-
rors?* Our key observation is that the corrupted-text setting
naturally produces preference pairs. If a model answers in-
correctly and its prediction appears in the corrupted auxil-
iary text, then the model has likely followed the mislead-
ing text. The ground-truth answer can be used as the pre-
ferred response, and the model’s text-following prediction
can be used as the dispreferred response. This gives a high-
precision preference signal without manual labeling.

We instantiate this idea with Direct Preference Opti-
mization (DPO) [11]. Unlike online reinforcement learning
methods such as PPO or GRPO [12], DPO directly opti-

mizes pairwise preferences. This is well aligned with our setting: the contrastive pair is already available as {image-grounded answer, text-following wrong answer}. Prior work already showed that supervised fine-tuning (SFT) with text augmentation can mitigate blind faith in text [3]. Our goal is not to deny that result, but to ask whether a more targeted preference signal can address the same failure mode. Instead of training on a broad mixture of text-only, original VQA, match, corruption, and irrelevance samples, we mine the specific cases where the model follows corrupted text and directly contrast that answer with the ground truth. We compare DPO against GRPO-style sampled-answer optimization, auxiliary SFT preservation, model transfer, and dataset transfer.

Our study makes three empirical findings. First, on DocVQA with Qwen3-VL-8B, DPO sharply improves corrupted-text robustness while reducing the rate of text-copying among remaining errors. Second, broadening DPO from corrupted-only text-following errors to multi-condition error-mined pairs yields the strongest result, suggesting that condition-specific error mining can improve stability. Third, the method has an important boundary condition: it works best when the base VLM already has enough visual/OCR competence. Qwen-family models benefit substantially, while LLaVA-Next models under our prompt and template show weak DocVQA base performance and limited benefit.

We summarize our contributions as follows:

- We define an automatic text-following error mining procedure for corrupted-text VQA, requiring both incorrectness and overlap between the wrong prediction and corrupted auxiliary text.
- We show that DPO on these mined pairs substantially reduces text bias on DocVQA and transfers directionally to VQAv2 and Qwen2-VL.
- We position DPO relative to the SFT mitigation studied by Deng et al. [3], and compare it with GRPO, auxiliary SFT preservation, multi-condition error mining, and model transfer.

2. Related Work

Text bias in VLMs. VLMs are trained to fuse visual and textual inputs, but fusion can become asymmetric when textual context is easier to exploit than visual evidence. The most directly related work is Words or Vision [3], which introduces corrupted textual variations and shows that VLMs may trust text over vision when the two conflict. Our work follows this diagnostic setting but focuses on mitigation through preference optimization. Rather than only measuring the degradation under corrupted text, we use the model’s own corrupted-text failures as training signal. Deng et al. [3] also study SFT with text augmentation and show that increasing SFT data reduces reliance on corrupted or

irrelevant text while preserving the ability to use matched text. They further note that text-only data helps prevent models from rejecting text indiscriminately. We build on this insight: the goal is conditional text use, not blanket text removal. Our contribution is to replace broad SFT augmentation with model-specific preference pairs mined from actual text-following failures. Other recent conflict-oriented evaluations similarly study how VLMs behave when image and text cues disagree. Mixed Signals [10] constructs mismatched image-text pairs across domains and studies model biases under conflicting signals. These works motivate our focus on conflict resolution rather than ordinary VQA accuracy alone. In contrast to benchmark-only studies, we explicitly turn conflict errors into training pairs.

Document and visual question answering. DocVQA [7] evaluates question answering over document images, requiring OCR, layout understanding, and reasoning over visual text. VQAv2 [4] evaluates open-ended visual question answering with reduced language priors compared to earlier VQA datasets. We use DocVQA as the main testbed because conflict between visual document evidence and auxiliary text is central to the task. VQAv2 is used as a transfer check to test whether the mitigation is specific to document images. Text-centric VQA datasets such as TextVQA [13], ST-VQA [2], OCR-VQA [9], and InfographicVQA [8] emphasize reading and reasoning over text inside images. These tasks are related but not identical to our corrupted-auxiliary-text setting. In TextVQA, scene text is often the correct visual evidence; in our DocVQA conflict setting, auxiliary text can be an external shortcut that contradicts the image. This distinction explains why we report TextVQA as a boundary check rather than as the main benchmark.

Preference optimization. Preference optimization has become a common alternative to supervised fine-tuning for aligning generative models. DPO [11] directly optimizes a policy from chosen/rejected pairs without training an explicit reward model or running online RL. This makes it attractive for our setting because corrupted-text evaluation yields natural preference pairs. GRPO [12] is an online reinforcement learning method that estimates a group-relative baseline from multiple sampled responses. We compare DPO with GRPO-style optimization and find that free-form sampled-answer rewards are less effective for this failure mode. Several recent works adapt preference optimization to multimodal models. mDPO [14] argues that standard multimodal DPO can suffer from unconditional preference learning, where the image condition is underused. Re-Align [16] constructs dual preference data with retrieval to incorporate visual preference signals. Task Preference Optimization [17] uses task-specific visual supervision to

improve multimodal perception, and on-policy DPO has been explored for hallucination mitigation [18]. Our work is complementary: rather than building generic human or retrieval preferences, we mine preference pairs from a specific modality-conflict failure mode.

Modern VLMs. Qwen2-VL [15] and Qwen2.5-VL [1] emphasize dynamic-resolution visual processing and strong document/OCR capability. LLaVA-style models [5, 6] provide a widely used visual instruction tuning framework. Our transfer experiments suggest that these model families differ substantially in the base competence needed for text-bias correction.

3. Method

3.1. Problem Setup

Each example consists of an image x , a question q , auxiliary text t , and a ground-truth answer y . We consider three main textual conditions. In the *match* condition, the auxiliary text is consistent with the image. In the *irrelevant* condition, the text is unrelated. In the *corrupted* condition, the text contains a plausible but wrong answer. The desired behavior is not to ignore text unconditionally. Instead, the model should use text when it is reliable and fall back to visual evidence when text conflicts with the image.

3.2. Mining Text-Following Errors

We first evaluate a base VLM on corrupted examples. Let $\hat{y}$ be the model prediction. We define a *text-following error* when all of the following hold:

1. the model prediction is incorrect;
2. the prediction is non-empty and does not match the ground truth;
3. the prediction appears in the corrupted auxiliary text.

For the third condition, we use normalized string matching. For multi-token answers, we also accept high token overlap with the auxiliary text. This filtering is important: an arbitrary wrong answer is not necessarily a text-bias error, but an incorrect prediction copied from corrupted text is a much stronger signal.

For each mined error, we construct a DPO pair:

$$y^+ = y, \quad y^- = \hat{y}, \quad (1)$$

where y^+ is the image-grounded answer and y^- is the model’s text-following wrong answer. We call this dataset *text-following span DPO*.

3.3. DPO Objective

Given a policy π_θ and a reference policy π_{ref} , DPO optimizes the preference that y^+ should be more likely than y^-

under the prompt (x, q, t) :

$$\mathcal{L}_{\text{DPO}} = -\log \sigma \left(\beta \left[\log \pi_\theta(y^+ | x, q, t) - \log \pi_{\text{ref}}(y^+ | x, q, t) - \log \pi_\theta(y^- | x, q, t) + \log \pi_{\text{ref}}(y^- | x, q, t) \right] \right). \quad (2)$$

In our setting, this objective directly encodes the desired correction: the image-grounded answer should be preferred over the corrupted-text answer.

3.4. Multi-Condition Error-Mined DPO

Text-following span DPO uses only corrupted examples. To reduce drift and improve stability, we also test a multi-condition extension. We mine errors from corrupted, match, irrelevant, and no-text evaluations. For corrupted examples, we keep the strict text-following filter: the wrong prediction must appear in corrupted auxiliary text. For match, irrelevant, and no-text examples, we use a broader condition-specific error filter: the prediction must be incorrect, non-empty, and distinct from the ground truth.

We assign lower weights to non-corrupted errors:

$$\begin{aligned} w_{\text{corrupted}} &= 1.0, & w_{\text{match}} &= 0.25, \\ w_{\text{irrelevant}} &= 0.5, & w_{\text{no-text}} &= 0.5. \end{aligned} \quad (3)$$

This preserves the central text-bias signal while adding weak regularization from other conditions.

3.5. GRPO-Style Baselines

We also test GRPO-style optimization. For each prompt, the model samples multiple candidate answers. A reward function gives positive reward for matching the chosen answer and penalties for matching the rejected answer, misleading spans mined from corrupted text, or any answer copied from the auxiliary text. The reward also mildly encourages short answer-like completions. This tests whether online sampled-answer optimization can discover the same correction signal without explicit pairwise DPO.

4. Experiments

4.1. Metrics

We report three complementary metrics. *Corrupted accuracy* measures task performance when auxiliary text is misleading. *Incorrect aux-hit* measures the fraction of incorrect predictions that appear in the corrupted auxiliary text. Lower incorrect aux-hit indicates less text-following among remaining errors, but it is not sufficient by itself: a model could reduce aux-hit by hallucinating unrelated answers. Therefore, we interpret it together with corrupted accuracy and preservation on match/irrelevant conditions. We also report corrected/regressed counts relative to the base model when available.

Table 1. DocVQA corrupted held-out results with Qwen3-VL-8B.

Method	Acc.	Incorrect aux-hit	Corr./Reg.
Base	0.715	89.5%	–
TF-span DPO, r8	0.900	55.0%	37 / 0
TF-span DPO, r16	0.915	47.1%	40 / 0
Multi-cond. error DPO	0.920	43.8%	41 / 0

4.2. Implementation Details

Unless otherwise stated, we train Qwen3-VL-8B-Instruct with LoRA adapters. The main DocVQA split uses 800 corrupted examples for mining and a disjoint held-out corrupted set of 200 examples for evaluation. The text-following span dataset is built only from base-model failures satisfying the corrupted-text overlap filter. For the initial span-DPO run, we use LoRA rank 8, LoRA alpha 16, learning rate 10^{-5} , DPO $\beta = 0.1$, and 700 DPO steps. The stronger rank-16 runs use LoRA alpha 32 with the same learning rate and number of steps. The multi-condition error-mined run uses 273 total preference rows, split into 245 train and 28 test rows, with source counts of 200 corrupted, 7 match, 33 irrelevant, and 33 no-text errors. It trains for 900 DPO steps with LoRA rank 16, alpha 32, learning rate 10^{-5} , and $\beta = 0.1$.

All reported transfer experiments use the same evaluation code and the same model family-specific prompt template used during mining. This matters because our LLaVA-Next experiments show strong sensitivity to task and template compatibility. We therefore interpret LLaVA results as boundary evidence rather than as a claim that the method is model-agnostic.

4.3. DocVQA Main Results

DocVQA is our main testbed. We use Qwen3-VL-8B-Instruct as the primary model and evaluate on a held-out corrupted split. Table 1 shows that text-following span DPO substantially improves both accuracy and text-bias behavior. Increasing LoRA rank from 8 to 16 improves the result further, while the multi-condition error-mined variant gives the best overall performance.

The multi-condition result suggests that the highest-performing variant is not corrupted-only. However, the corrupted filter remains essential: corrupted rows still require the prediction to be copied from corrupted auxiliary text. The additional match, irrelevant, and no-text errors act as weak condition-specific regularizers rather than replacing the text-following signal.

4.4. Hyperparameter and Construction Ablations

We run a small 2×2 grid over DPO β and LoRA rank on the same DocVQA split. Table 2 shows that increasing LoRA rank from 8 to 16 is more useful than increasing β from 0.1

Table 2. DocVQA DPO hyperparameter grid. Higher LoRA capacity matters more than β in this setting.

β	Rank	Acc.	Aux-hit	Corr./Reg.	Match/Irr.
0.1	8	0.900	55.0%	37 / 0	1.00 / 0.96
0.2	8	0.900	55.0%	37 / 0	0.98 / 0.91
0.1	16	0.915	47.1%	40 / 0	0.98 / 0.91
0.2	16	0.915	47.1%	40 / 0	0.98 / 0.91

Table 3. GRPO-style optimization does not improve over DPO on DocVQA.

Variant	Init	Acc.	Incorrect aux-hit
Base	–	0.715	89.5%
GRPO-only, 100 steps	base	0.710	89.7%
GRPO-only, 300 steps	base	0.705	89.8%
GRPO-only, dense reward	base	0.705	89.8%
TF-span DPO	base	0.900	55.0%
DPO $\rightarrow$ GRPO	DPO	0.900	60.0%

to 0.2. Rank 16 improves corrupted accuracy from 0.900 to 0.915 and reduces incorrect aux-hit from 55.0% to 47.1%, while changing β does not change the corrupted metric in this grid.

We also test whether simply duplicating the same corrupted preference pair under multiple prompt conditions is sufficient. Prompt-augmented counterfactual DPO expands each mined corrupted failure into corrupted, no-text, match, and irrelevant prompt wrappers, but keeps the same rejected answer. This ties plain rank-16 DPO at 0.915 corrupted accuracy and 47.1% incorrect aux-hit. By contrast, multi-condition error-mined DPO uses the actual failed prediction from each condition as the rejected answer, improving to 0.920 accuracy and 43.8% incorrect aux-hit. This suggests that condition-specific rejected answers are more useful than prompt augmentation alone.

4.5. DPO versus GRPO

We compare DPO with GRPO-only and DPO-to-GRPO variants on the same disjoint DocVQA split. GRPO samples multiple answers per prompt and optimizes a reward that encourages correctness and penalizes copying corrupted text. Despite this denser reward, GRPO does not move the held-out greedy behavior meaningfully. Table 3 shows that DPO is the effective intervention in our setting.

DPO-to-GRPO changes only a few held-out predictions relative to DPO and does not improve accuracy. This supports our hypothesis: when high-precision chosen/rejected pairs are available, direct pairwise optimization provides a clearer learning signal than free-form sampled-answer rewards.

Table 4. VQAv2 transfer results with Qwen3-VL-8B.

Variant	Corr. Acc.	Aux-hit	Match	Irr.
Base	0.545	65.9%	0.92	0.77
Span DPO	0.645	40.8%	0.90	0.75
Span DPO + SFT	0.635	42.5%	0.90	0.75

4.6. SFT: Prior Mitigation and Our Ablation

The original Words or Vision study already evaluates SFT with text augmentation as a mitigation strategy [3]. That SFT setting uses a broader training mixture, including original VQA, text-only, match, corruption, and irrelevance samples, and shows that SFT can reduce text reliance while maintaining the ability to benefit from matched text. This is an important baseline and motivates our own mitigation study.

Our experiment is different. We do not reproduce the full augmented SFT recipe from Deng et al. [3]. Instead, we test a narrower question: after constructing text-following DPO pairs, can an auxiliary SFT preservation loss on match/irrelevant examples further stabilize the model? This is a preservation ablation, not a claim that SFT in general is ineffective. SFT is plausible because it directly teaches the model to output correct answers. However, the central error in our setting is not simply that the model does not know the answer; it is that the model prefers a misleading textual answer over visual evidence. SFT provides only a positive target, while DPO explicitly contrasts the image-grounded answer against the text-following wrong answer.

Our SFT experiment uses auxiliary preservation data from match and irrelevant conditions and mixes it with span DPO. As shown in Table 4, this does not improve over pure DPO on VQAv2: corrupted accuracy decreases from 0.645 to 0.635, incorrect aux-hit increases from 40.8% to 42.5%, and match/irrelevant preservation remains unchanged. Thus, in our experiments, auxiliary SFT preservation acts as a negative ablation, while the prior full SFT augmentation result remains a valid mitigation baseline from the original work.

4.7. VQAv2 Transfer Check

We repeat the disjoint split protocol on VQAv2. The effect is positive but weaker than on DocVQA. As shown in Table 4, span DPO improves corrupted accuracy by 10 points and reduces incorrect aux-hit, but match and irrelevant preservation drop slightly. Adding auxiliary SFT preservation does not recover this drop and slightly worsens corrupted performance.

4.8. TextVQA Boundary Check

We also test whether a DocVQA-trained span DPO adapter transfers to TextVQA. TextVQA is useful because it is text-

Table 5. TextVQA question-type matched answer-hint transfer. DocVQA DPO transfers weakly to Qwen models but not reliably to LLaVA-Next.

Model	Base Acc.	DPO Acc.	Base aux-hit	DPO aux-hit
Qwen3-VL-8B	0.821	0.827	11.5%	8.3%
Qwen2-VL-7B	0.832	0.845	23.1%	8.3%
LLaVA-N-7B	0.535	0.443	50.0%	62.0%
LLaVA-N-13B	0.612	0.602	35.7%	28.8%

Table 6. Model transfer on DocVQA. DPO helps Qwen2-VL but not LLaVA-Next under our setting.

Model	Setting	Acc.	Aux-hit	Corr./Reg.
Qwen2-VL-7B	base	0.570	93.0%	–
Qwen2-VL-7B	span DPO	0.775	86.7%	41 / 0
LLaVA-N-7B	base	0.105	90.5%	–
LLaVA-N-7B	span DPO	0.100	91.7%	1 / 2
LLaVA-N-13B	base	0.115	92.7%	–
LLaVA-N-13B	span DPO	0.130	95.4%	3 / 0

centric, but it differs from DocVQA conflict: scene text is often the correct visual evidence rather than an external auxiliary shortcut. We therefore construct several TextVQA corruption variants, including same-image OCR distractors, cross-sample answers, and question-type matched answer hints.

Table 5 shows the strongest TextVQA conflict variant, question-type matched answer hints, on 200 examples. For Qwen3-VL, the base model already has low incorrect aux-hit under corrupted TextVQA hints, and DocVQA DPO gives only a small gain. Qwen2-VL shows a similar but slightly stronger positive direction. LLaVA-Next behaves differently: the conflict is stronger, but DocVQA DPO does not reliably fix it and can worsen corrupted accuracy.

These results sharpen the scope of our claim. The method is not a universal accuracy booster for all text-heavy VQA tasks. It is most effective when the base model actually exhibits text-following failures under corrupted auxiliary text and already has enough visual/OCR competence to answer from the image.

4.9. Model Transfer

We test whether the same mitigation extends beyond Qwen3-VL. Table 6 summarizes Qwen2-VL-7B and LLaVA-Next results on DocVQA. Qwen2-VL benefits substantially, improving from 0.570 to 0.775 corrupted accuracy. LLaVA-Next, however, has very low base DocVQA accuracy under our prompt/template, and DPO provides little benefit.

These results show that the method is not simply a universal post-hoc fix. It requires a base model that can already solve the task visually; DPO then teaches the model to resist

412 misleading text.

413 5. Analysis and Discussion

414 5.1. Why DPO Fits This Problem

415 The corrupted-text setting gives unusually clean preference
416 pairs. For many failures, the model’s wrong answer is di-
417 rectly copied from the corrupted auxiliary text. Thus the
418 comparison is explicit: the image-grounded answer should
419 be preferred over the text-following wrong answer. DPO
420 optimizes exactly this comparison. This complements,
421 rather than invalidates, the SFT mitigation in Words or Vi-
422 sion [3]. SFT with text augmentation teaches the model
423 from a balanced mixture of cases and can reduce broad
424 text reliance. Our DPO variant instead uses a narrower but
425 sharper signal: for each mined failure, it tells the model
426 which specific text-derived answer should lose to the visual
427 answer. In contrast, GRPO must first sample useful candi-
428 date answers and then rely on a hand-designed reward. If
429 the sampled candidates rarely contain the correct answer, or
430 if most candidates are similar text-following errors, the re-
431 lative reward signal is weak. This explains why GRPO logs
432 can show diagnostic reward events while greedy held-out
433 predictions barely change.

434 5.2. Why Not Simply Ignore the Text?

435 Ignoring auxiliary text is an appealing but incomplete base-
436 line. It would avoid some corrupted-text failures, but it
437 does not solve the actual decision problem faced by a VLM.
438 At inference time, the model is not given a label saying
439 whether the text is matched, irrelevant, or corrupted. If the
440 model learns to discard text wholesale, it can lose useful
441 information in match cases and may fail to exploit reliable
442 OCR or retrieved context.

443 Our objective is therefore not “use less text” but “use text
444 conditionally”. The model should keep using textual con-
445 text when it is visually supported and reject it when it con-
446 flicts with the image. DPO is appropriate because it changes
447 the model’s preference in exactly the conflict cases we can
448 identify:

image-grounded answer $\succ$ text-following wrong answer.
(4)

449 This is different from input ablation or prompt-level in-
450 struction to ignore text. The latter can reduce exposure to
451 misleading text, but it does not teach the model which an-
452 swer should be preferred when both visual evidence and
453 misleading text are present. Our use of match and irrele-
454 vant preservation metrics is intended to check precisely this
455 point: a successful method should improve corrupted ro-
456 bustness without collapsing into text avoidance.

5.3. Interpreting Incorrect Aux-Hit

Incorrect aux-hit is useful but should not be optimized
alone. A lower value means fewer remaining errors are
copied from corrupted text, but it does not guarantee bet-
ter visual grounding. For example, a model could avoid the
corrupted text yet hallucinate an unrelated answer. There-
fore, successful text-bias correction should satisfy three
conditions:

1. corrupted accuracy should increase;
2. incorrect aux-hit should decrease;
3. match and irrelevant performance should be preserved.

Our main DocVQA results satisfy these conditions. Some
negative ablations do not: they reduce neither error copy-
ing nor accuracy, or they preserve match performance while
failing on corrupted examples.

5.4. Why Multi-Condition Error Mining Helps

Text-following span DPO is the core filtering strategy, but
the best result comes from multi-condition error-mined
DPO. This does not mean that unfiltered errors are suffi-
cient. Rather, corrupted errors remain strictly filtered for
text-following behavior, while other conditions contribute
lower-weight stability signals. This helps prevent the model
from over-specializing to the corrupted prompt format. The
gain is modest because non-corrupted failures are sparse in
DocVQA, but it consistently improves the main held-out
metric.

5.5. Boundary Conditions

Our transfer experiments reveal a clear boundary condition.
The method works when the base VLM already has suffi-
cient visual and OCR competence. Qwen-family models
satisfy this condition on DocVQA and improve substan-
tially. LLaVA-Next models, under our prompt/template,
have very low base DocVQA accuracy. In this regime, DPO
cannot create document-reading ability from scratch; it can
only alter preferences among answers the model can already
represent.

TextVQA experiments lead to a related conclusion.
DocVQA-trained DPO transfers weakly to TextVQA and
does not reliably fix LLaVA-specific TextVQA conflict fail-
ures. Even after mining larger LLaVA-specific TextVQA
DPO data and adding strict+preserve pairs, corrupted per-
formance does not improve. This suggests that TextVQA
conflict involves different OCR and answer-format dy-
namics than DocVQA. Therefore, we treat TextVQA and
LLaVA results as boundary analyses rather than main suc-
cess claims.

5.6. Limitations

Our study has several limitations. First, the text-following
filter is an operational definition, not a causal proof. If a
wrong answer appears in corrupted text, the model likely

followed the text, but accidental overlap is possible. Second, most experiments use a single held-out split and a limited number of seeds due to compute constraints. Third, our GRPO implementation uses free-form sampled answers; a candidate-constrained GRPO-style objective may be a stronger comparison. Finally, the method assumes ground-truth answers are available for preference construction. This is suitable for benchmark adaptation and supervised mitigation, but not directly for fully unlabeled deployment.

6. Conclusion

We studied preference optimization as a mitigation for VLM text bias under corrupted auxiliary text. By mining text-following errors and training with DPO, we convert a diagnostic failure mode into a direct preference-learning signal. On DocVQA, this substantially improves corrupted-text accuracy and reduces the rate at which remaining errors copy misleading text. The strongest variant, multi-condition error-mined DPO, combines strict corrupted-error filtering with lower-weight errors from match, irrelevant, and no-text conditions. GRPO-style sampled-answer optimization and preservation SFT do not improve over DPO in our experiments. These comparisons also clarify why simply ignoring text is not the desired solution: the target behavior is conditional trust, not blanket text removal. The results support a precise conclusion: DPO is effective when the base VLM already has visual/OCR competence and the preference pairs accurately capture text-following failures. When this condition is not met, as in our LLaVA-Next DocVQA setting, preference optimization alone is insufficient.

References

- [1] Shuai Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, Sibao Song, Kai Dang, Peng Wang, Shijie Wang, Jun Tang, et al. Qwen2.5-vl technical report. *arXiv preprint arXiv:2502.13923*, 2025. 3
- [2] Ali Furkan Biten, Ruben Tito, Andres Mafla, Lluís Gomez, Marçal Rusiñol, Ernest Valveny, C. V. Jawahar, and Dimosthenis Karatzas. Scene text visual question answering. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2019. 2
- [3] Ailin Deng, Tri Cao, Zhirui Chen, and Bryan Hooi. Words or vision: Do vision-language models have blind faith in text? In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025. 1, 2, 5, 6
- [4] Yash Goyal, Tejas Khot, Douglas Summers-Stay, Dhruv Batra, and Devi Parikh. Making the v in vqa matter: Elevating the role of image understanding in visual question answering. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, 2017. 2
- [5] Haotian Liu, Chunyuan Li, Qingyang Wu, and Yong Jae Lee. Visual instruction tuning. *Advances in Neural Information Processing Systems*, 2023. 3 616
- [6] Haotian Liu, Chunyuan Li, Yuheng Li, Bo Li, Yuanhan Zhang, Sheng Shen, and Yong Jae Lee. Llava-next: Improved reasoning, ocr, and world knowledge. *LLaVA project blog*, 2024. 3 559 560 561 562
- [7] Minesh Mathew, Dimosthenis Karatzas, and C. V. Jawahar. Docvqa: A dataset for vqa on document images. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, 2021. 2 563 564 565 566
- [8] Minesh Mathew, Viraj Bagal, Ruben Tito, Dimosthenis Karatzas, Ernest Valveny, and C. V. Jawahar. Infograph-icvqa. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, 2022. 2 567 568 569 570
- [9] Anand Mishra, Shashank Shekhar, Ajeet Kumar Singh, and Anirban Chakraborty. Ocr-vqa: Visual question answering by reading text in images. In *International Conference on Document Analysis and Recognition*, 2019. 2 571 572 573 574
- [10] Pouya Pezeshkpour, Moin Amnaseri, and Estevam Hruschka. Mixed signals: Decoding vlms' reasoning and underlying bias in vision-language conflict. In *Findings of the Association for Computational Linguistics: EMNLP 2025*, pages 24833–24848, 2025. 2 575 576 577 578 579
- [11] Rafael Rafailov, Archit Sharma, Eric Mitchell, Stefano Ermon, Christopher D. Manning, and Chelsea Finn. Direct preference optimization: Your language model is secretly a reward model. In *Advances in Neural Information Processing Systems*, 2023. 1, 2 580 581 582 583 584
- [12] Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, Y. K. Li, Y. Wu, and Daya Guo. Deepseekmath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*, 2024. 1, 2 585 586 587 588 589
- [13] Amanpreet Singh, Vivek Natarajan, Meet Shah, Yu Jiang, Xinlei Chen, Dhruv Batra, Devi Parikh, and Marcus Rohrbach. Towards vqa models that can read. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2019. 2 590 591 592 593 594
- [14] Fei Wang, Wenxuan Zhou, James Y. Huang, Nan Xu, Sheng Zhang, Hoifung Poon, and Muhao Chen. mdpo: Conditional preference optimization for multimodal large language models. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, 2024. 2 595 596 597 598 599
- [15] Peng Wang, Shuai Bai, Sinan Tan, Shijie Wang, Zhihao Fan, Jinze Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, et al. Qwen2-vl: Enhancing vision-language model's perception of the world at any resolution. *arXiv preprint arXiv:2409.12191*, 2024. 3 600 601 602 603 604
- [16] Shuo Xing, Yuping Wang, Peiran Li, Ruizheng Bai, Yueqi Wang, Chengxuan Qian, Huaxiu Yao, and Zhengzhong Tu. Re-align: Aligning vision language models via retrieval-augmented direct preference optimization. *arXiv preprint arXiv:2502.13146*, 2025. 2 605 606 607 608 609
- [17] Ziang Yan, Zhilin Li, Yinan He, Chenting Wang, Kunchang Li, Xinhao Li, Xiangyu Zeng, Zilei Wang, Yali Wang, Yu Qiao, Limin Wang, and Yi Wang. Task preference optimization: Improving multimodal large language models with vision task alignment. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025. 2 610 611 612 613 614 615

- 617 [18] Zhihe Yang, Xufang Luo, Dongqi Han, Yunjian Xu, and
618 Dongsheng Li. Mitigating hallucinations in large vision-
619 language models via dpo: On-policy data hold the key. In
620 *Proceedings of the IEEE/CVF Conference on Computer Vi-
sion and Pattern Recognition*, 2025. 3

621